

DESIGN AND ANALYSIS OF ALGORITHM

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RGCET

HOMOGENOUS RECURRENCES

We begin our study of the technique of the characteristics equation with the resolution of homogeneous linear recurrence with constant co-efficient, i.e. the recurrence of the form,

$$a_0 t_n + a_1 t_{n-1} + \dots + a_k t_{n-k} = 0$$

where t is the values we are looking for.

The values of t_i on k values of i

(usually $0 \leq i' \leq k-1$ (or) $0 \leq i \leq (k)$) are needed to determine the sequence.

The initial condition will be considered later. The equation typically has infinitely many solutions.

The recurrence is,

Linear because it does not contain terms of the t_{n-i} , t_{n-j} , t^2_{n-i} , and so on.

Homogeneous because the linear combination of the t_{n-i} , t_{n-j} , t^2_{n-i} is equal to zero.

With the co-efficient because the a_i constant consider for instance our non familiar recurrence for the fibonacci sequence.

$$f_n = f_{n-1} + f_{n-2}.$$

Therefore, the fibonacci sequence corresponds to a homogeneous linear recurrence with constant co-efficient $k=2$, $q_0 = 1$, $q_1 = a_1 = -1$

This is true because,

$$a_0 t^n + a_1 t^{n-1} + \dots + a_k t^{n-k}$$

$$= a_0 (c f^n + d g^n) + a_1 (c f^{n-1} + d g^{n-1}) + \dots + a_k (c f^{n-k} + d g^{n-k})$$

$$= c (a_0 f^n + a_1 f^{n-1} + \dots + a_k f^{n-k}) + d (a_0 g^n + a_1 g^{n-1} + \dots + a_k g^{n-k})$$

$$= c * 0 + d * 0 = 0.$$

1) Fibonacci, consider the recurrence

$$f(n) = \begin{cases} n & \text{if } n=0 \text{ or } n=1 \\ f_{n-1} + f_{n-2} & \text{otherwise} \end{cases}$$

We rewrite the recurrence as,

$$f_n - f_{n-1} - f_{n-2} = 0$$

The characteristic polynomial is,

$$x^2 - x - 1 = 0$$

The roots are,

$$x = \frac{-(-1) \pm \sqrt{(-1)^2 + 4}}{2}$$

$$= \frac{1 \pm \sqrt{1+4}}{2}$$

$$= \frac{1 \pm \sqrt{5}}{2}$$

$$r_1 = \frac{1+\sqrt{5}}{2} \quad \text{and} \quad r_2 = \frac{1-\sqrt{5}}{2}$$

The general solutions are,

$$f_n = C_1 r_1^n + C_2 r_2^n$$

When $n=0$, $f_0 = C_1 + C_2 = 0$

When $n=1$, $f_1 = C_1 r_1 + C_2 r_2 = 1$

$$C_1 + C_2 = 0 \quad \text{--- (1)}$$

$$C_1 r_1 + C_2 r_2 = 1 \quad \text{--- (2)}$$

From equation (1)

$$C_1 = -C_2$$

Substitute r_1 and r_2 values

$$C_2 = \frac{1-\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = 1$$

$$C_2 = \frac{1-\sqrt{5} - 1 - \sqrt{5}}{2} = 1$$

$$= -\frac{C_2 + 2\sqrt{5}}{2} = 1$$

$$\sqrt{5} C_2 = 1$$

$$C_1 = \frac{1}{\sqrt{5}}$$

$$C_2 = \frac{-1}{\sqrt{5}}$$

Thus,

$$f_n = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n + \frac{-1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^n$$

$$= \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^n - \frac{1-\sqrt{5}}{2}^n$$

INHOMOGENEOUS RECURRENCE

* The solution of a linear recurrence with constant co-efficient becomes more difficult when the recurrence is not homogeneous, that is when the linear combination is not equal to zero.

* Consider the following recurrence

$$a_0 t_n + a_1 t_{n-1} + \dots + a_k t_{n-k} = b^n p(n)$$

* The left hand side is the same as before, (homogeneous) but on the right-hand side we have $b^n p(n)$, where

→ b is a constant

→ $p(n)$ is a polynomial in ' n ' of degree ' d '.

Example

Consider the recurrence

$$t_n - 2t_{n-1} = 3^n \quad \text{--- (A)}$$

In this case, $b=3$, $p(n)=1$, degree = 0;

The characteristic polynomial is

$$(x-2)(x-3)=0.$$

The roots are, $r_1=2$, $r_2=3$

The general solution

$$\begin{aligned} t_n &= C_1 r_1^n + C_2 r_2^n \\ t_n &= C_1 2^n + C_2 3^n \quad \text{--- (1)} \end{aligned}$$

When $n=0$, $C_1 + C_2 = t_0$ — (2)

When $n=1$, $2C_1 + 3C_2 = t_1$ — (3)

Sub $n=1$ in equation (A)

$$t_1 = 2t_0 = 3$$

$$t_1 = 3 + 2t_0$$

Substitute t_1 in eqⁿ (3)

$$(2) + 2 \rightarrow 2C_1 + 2C_2 = 2t_0$$

$$2C_1 + 3C_2 = (3 + 2t_0)$$

$$-C_2 = -3$$

$$C_2 = 3$$

Sub $C_2 = 3$ in eqⁿ (2)

$$C_1 + C_2 = t_0$$

$$C_1 + 3 = t_0$$

$$C_1 = t_0 - 3$$

$$\text{Therefore } t_n = (t_0 - 3)2^n + 3 \cdot 3^n$$

$$= \max [O((t_0 - 3)2^n), O[3 \cdot 3^n]]$$

$$= \max [O(2^n), O(3^n)] \text{ constants}$$

$$= O[3^n]$$

STRASSON'S MATRIX

- * Let A and B be the $2n \times n$ Matrix. The product matrix $C = AB$ is calculated by using the formula, $C(i,j) = A(i,k) B(k,j)$ for all i and j between 1 and n .
- * The time complexity for the matrix multiplication is $O(n^3)$
- * Divide and conquer method suggest another way to compute the product of $n \times n$ matrix.
- * We assume that N is a power of 2. In the case columns of zero can be added to both A and B .
So that the resulting dimension are the power of two.
- * If $n=2$ then the following formula as a computed using a matrix multiplication operation for the elements of A & B .
- * If $n > 2$, then the elements are partitioned into sub matrix $n/2 \times n/2 \dots$. Since ' n ' is a power of 2 these product can be recursively computed using the same formula. This algorithm will continue applying itself to smaller sub matrix until ' N ' become suitable small ($n=2$) so that the product is computed directly.

The formula are,

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} * \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$$C_{11} = A_{11} B_{11} + A_{12} B_{21}$$

$$C_{12} = A_{11} B_{12} + A_{12} B_{22}$$

$$C_{21} = A_{21} B_{11} + A_{22} B_{21}$$

$$C_{22} = A_{21} B_{12} + A_{22} B_{22}$$

For Ex:

$$4 \times 4 = \begin{bmatrix} 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \\ 2 & 2 & 2 & 2 \end{bmatrix} * \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

The divide and conquer method

$$\left[\begin{array}{c|c} \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} & \begin{bmatrix} 2 & 2 \end{bmatrix} \\ \hline \begin{bmatrix} 2 & 2 \end{bmatrix} & \begin{bmatrix} 2 & 2 \end{bmatrix} \end{array} \right] * \left[\begin{array}{c|c} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \\ \hline \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \end{array} \right] = \left[\begin{array}{c|c} \begin{bmatrix} 4 & 4 \end{bmatrix} & \begin{bmatrix} 4 & 4 \end{bmatrix} \\ \hline \begin{bmatrix} 4 & 4 \end{bmatrix} & \begin{bmatrix} 4 & 4 \end{bmatrix} \end{array} \right]$$

* To compute AB using the equation we need to perform 8 multiplication of $n/2 \times n/2$ matrix and from 4 addition of $n/2 \times n/2$ matrix.

* $C_{i,j}$ are computed using the formula in equation $\rightarrow 4$

* As can be sum P, q, R, S, T, U and V can be computed using 7 matrix multiplication and 10 addition or subtraction.

* The C_{ij} are required addition & subtraction.

$$T(n) = \begin{cases} b & n \leq 2 \\ 7T(n/2) + an^2 & n > 2 \end{cases} \quad \text{where } a, b \text{ are constant}$$

Finally we get $T(n) = O(n^{\log_2 7})$

Example

$$\begin{vmatrix} 4 & 4 \\ 4 & 4 \end{vmatrix} * \begin{vmatrix} 4 & 4 \\ 4 & 4 \end{vmatrix}$$

$$P = (4 * 4) + (4 * 4) = 64$$

$$Q = (4 + 4) * 4 = 32$$

$$R = 4 * (4 - 4) = 0$$

$$S = 4 * (4 - 4) = 0$$

$$T = (4 + 4) * 4 = 32$$

$$U = (4 - 4) * (4 + 4) = 0$$

$$V = (4 - 4) * (4 + 4) = 0$$

$$C_{11} = (64 + 0 - 32 + 0) = 32$$

$$C_{12} = 0 + 32 = 32$$

$$C_{21} = 32 + 0 = 32$$

$$C_{22} = 64 + 0 - 32 + 0 = 32$$

So the answer $C(i, j)$ is $\begin{vmatrix} 32 & 32 \\ 32 & 32 \end{vmatrix}$

Since $n/2 \times n/2$ matrix can be added in

C_n for some constant C . The overall computing

time $T(n)$ of the resulting divide and conquer algorithm is given by the sequence.

$$T(n) = \begin{cases} b & n \leq 2 \text{ a \& b are} \\ 8T(n/2) + (n^2) & n > 2 \text{ constant} \end{cases}$$

That is $T(n) = O(n^3)$

Matrix multiplication are more expensive than the matrix addition $O(n^3)$, we can attempt to reformulate the equation for (i) so as to have fewer multiplications and possibly more addition.

→ Strassen has discovered a way to compute the (i) of equation (2) using only 7 multiplications and 18 addition or subtraction.

→ Strassen's formula are

$$P = (A_{11} + A_{12})(B_{11} + B_{22})$$

$$Q = (A_{12} + A_{22})B_{11}$$

$$R = A_{11}(B_{12} - B_{22})$$

$$S = A_{22}(B_{21} - B_{11})$$

$$T = (A_{11} + A_{12})B_{22}$$

$$U = (A_{21} - A_{11})(B_{11} + B_{12})$$

$$V = (A_{12} - A_{22})(B_{11} + B_{22})$$

$$C_{11} = P + S - T + V$$

$$C_{12} = R + T$$

$$C_{21} = Q + T$$

$$C_{22} = P + R - Q + V$$